

Supersensitive Sm^{2+} -Activated Al_2O_3 Thermometric Coatings for High-Resolution Multiple Temperature Read-Outs from Luminescence

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The introduction of additional functionalities to materials is exceptionally important as it opens new applications for them. Aluminum, one of the most abundant and important materials, is coated with luminescent Sm^{2+} -doped γ -aluminium oxide to impart thermometric functionality. Considering the potential industrial applications, two of the most widely used aluminum alloys, 6061 and 7075, are also coated. For this purpose, plasma electrolytic oxidation (PEO), an effective technique for producing hard ceramic coatings on various metal surfaces, is used. It is shown that thermometric coatings can be produced on aluminum in one-step process by adding the raw precursor to the electrolyte. The valence reduction of Ln^{3+} to Ln^{2+} is achieved during the PEO process. The intense and broad (orange to deep red) emission from the coating shows supersensitivity to temperature changes over the 100–648 K range. The temperature is obtained from the coating emission using i) the emission intensity ratio method, ii) emission lifetime, and iii) emission band position with sensitivities of $4.8\% \text{ K}^{-1}$, $1.2\% \text{ K}^{-1}$, and $8 \text{ cm}^{-1} \text{ K}^{-1}$, respectively. Several applications would benefit from the thermometric coating's excellent temperature resolution of 0.04 K and the choice of three temperature read-outs that facilitate the coating's use in different luminescence thermometry setups.

components and other sporting equipment, camera lenses, firearms, electrical fittings, and aluminum cans in the food industry.^[1–6] The chemical compositions and mechanical characteristics of these alloys are given in ref. [7]. Aluminum has a great affinity towards atmospheric oxygen and thin passivation layer of aluminum oxide forms spontaneously and rapidly on any exposed aluminum surface.^[8–10] This self-healing layer protects the metal from further oxidation and its thickness and properties can be enhanced and modified using different processes.

Aluminum oxide (Al_2O_3), an important coating material, is either grown from aluminum or deposited on the substrate of choice. Al_2O_3 is, after diamond, the second hardest material, susceptible to high temperatures, resistant to high mechanical stress, and chemically stable.^[11–13] Among the many stable and metastable phases (α , β , γ , η , θ , κ , δ , χ),^[14] $\gamma\text{-Al}_2\text{O}_3$ is particularly important in modern industries and has

1. Introduction

Aluminum is one of the most abundant and industrially important materials. It is frequently alloyed, which significantly improves its mechanical properties. Among the many alloys, 6061 and 7075 are commonly used in aeronautics, automotive and marine industries, construction, high-quality bicycle

no shortage of applications. As a multifunctional metal coating, it is used for the prevention of abrasive and thermal wear; it is also used as a catalyst in spacecraft, power supplies, electronics, optics, and other applications.^[15,16] Having a wide energy band-gap (5.4 eV), high refractive index (1.86), and a cutoff phonon energy of 750 cm^{-1} ,^[16,17] $\gamma\text{-Al}_2\text{O}_3$ is an excellent host material for facilitating lanthanide (Ln) or transition metal (TM) ion emission. By doping Al_2O_3 with Ln or TM ions, additional functionalities of alumina can be engineered, and new application areas may be opened.

Luminescence thermometry, a technique using luminescence as a temperature-responsive signal for measuring temperature, is based on the observation of luminescence changes induced by temperature fluctuations (such as the change in the emission peak energy and intensity, emission or excitation band shapes, excited-state lifetimes, and rise times, and polarization).^[18,19] Being essentially a semi-contact method, luminescence thermometry succeeds in environments where many other, if not all, thermometry methods cannot be applied or tend to fail. For example, luminescence thermometry shows extraordinary benefits when applied to temperature measurements in high-temperature environments for the development

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and safe use of chemical reactors, engines, and gas turbines.^[20] In order to overcome the applicability limitations of luminescent thermometry in terms of sensitivity and adaptability to existing thermometry setups, new luminescence probes and temperature read-out approaches need to be developed.

So far, luminescence thermometry commonly exploited Boltzmann-type luminescence intensity ratio (LIR) between thermally coupled energy levels of different trivalent Ln ions. However, relative sensitivity of the thermometry is limited by the energy gap between the levels, reaching maximum for Eu^{3+} ions whose $^5\text{D}_1$ and $^5\text{D}_0$ emissions utilized for LIR are separated by approximately 1750 cm^{-1} . Only recently, divalent Ln ions having both $4\text{f}5\text{d}$ - 4f and $4\text{f}4\text{f}$ emissions separated with the larger energy gap are considered promising to meet the higher sensitivity requirements. Having in mind that among Ln ions only samarium and europium are used in the divalent form in phosphors, the present research introduces Sm^{2+} -doped $\gamma\text{-Al}_2\text{O}_3$ as a luminescent coating with embedded thermometric functionality. To the best of our knowledge, this is the first report on luminescence thermometry with Ln^{2+} -doped $\gamma\text{-Al}_2\text{O}_3$ coatings. Plasma electrolytic oxidation (PEO) of an aluminum substrate was successfully developed and used for the fabrication of coatings in the present authors' previous studies.^[21–24] In addition, the valence reduction of Ln-ions in the host materials is usually accomplished by fabrication in a reducing atmosphere.^[25,26] However, the PEO processing conditions, with locally evolving high temperatures and pressures,^[27] lead to the reduction of Ln^{3+} to Ln^{2+} in $\gamma\text{-Al}_2\text{O}_3$ coatings.^[22–24,28,29] Owing to the energetically low $4\text{f}5\text{d}$ electronic configuration of the Sm^{2+} in the $\gamma\text{-Al}_2\text{O}_3$ host, Sm^{2+} emission can be excited by any wavelength shorter than 550 nm . It has been shown that the intense emission of Sm^{2+} is strongly temperature-dependent, making Sm^{2+} emission suitable for temperature sensing.^[30] In this study, the sensor coatings were prepared on 99.9% pure aluminum foil and on commercial 6061 and 7075 aluminum alloys as used in the automotive and aerospace industries. Since PEO coatings are directly grown on the substrate they can achieve far better adhesion to the substrate than the traditional compact luminescent temperature sensors, which leads to better thermal contact between the sensor and the measuring object.^[31,32]

2. Results and Discussion

The morphology, structure, photoluminescence (PL), and thermometric properties of Sm^{2+} -doped $\gamma\text{-Al}_2\text{O}_3$ coatings created by PEO using pure aluminum and 6061 and 7075 alloy substrates, respectively, showed no significant differences. Therefore, only the results for the Sm^{2+} -doped $\gamma\text{-Al}_2\text{O}_3$ coating deposited on the pure aluminum substrate are presented in the main text. The PL and thermometric properties of Sm^{2+} -doped $\gamma\text{-Al}_2\text{O}_3$ coatings created by PEO using 6061 and 7075 alloy substrates are presented in the Supporting Information (Figures S2–S4 and Table S2, Supporting Information).

The porous surface morphology with different-sized voids, typical for the coatings prepared by the PEO process (which includes a large number of sequential microscopic melting steps and rapid solidification)^[33] is visible on the top-view coating images presented in Figure 1a,b. The cross-sectional

scanning electron microscopy presented in Figure 1c shows a completely dense PEO coating with an average thickness of $\approx 5\text{ }\mu\text{m}$ uniformly distributed over the aluminum surface. The X-ray diffraction (XRD) pattern of the representative Sm^{2+} -doped pure alumina PEO coating is presented in Figure 1d. The XRD analysis revealed peaks at positions corresponding to the cubic γ -phase of Al_2O_3 , with no traces of the α - Al_2O_3 phase and with the most intense diffraction peaks corresponding to the aluminum substrate. One would expect a corundum-type α - Al_2O_3 structure, which is thermodynamically more stable under ambient conditions,^[14] to be deposited. One of the reasons for deposition of γ - Al_2O_3 could be the fact that cubic γ - Al_2O_3 crystal system grows in an epitaxial fashion on the top of cubic Al substrate. Also, the reason might be the coating porosity inherent for the violent PEO process, as it is known that the gamma phase always appears in the porous morphology.^[34] A representative Sm^{2+} -doped $\gamma\text{-Al}_2\text{O}_3$ coating was recorded under daylight conditions (Figure 1g), and under 473 nm laser irradiation in the dark, with i) a 550 nm high-pass filter mounted on the camera (Figure 1f,i) and ii) without the filter (Figure 1e,h). The orange emission color of the coating is clearly visible through the long-pass filter (550 nm) under 473 nm illumination.

The PL excitation spectrum recorded over the 250 – 550 nm range (emission monitored at $\lambda_{\text{em}} = 688\text{ nm}$) of the Sm^{2+} -doped $\gamma\text{-Al}_2\text{O}_3$ PEO coating is shown in Figure 2a. It shows a broad excitation band from the $^7\text{F}_1 \rightarrow 4\text{f}5\text{d}$ electronic transitions (note that the excitation of Sm^{2+} starts from levels with $J \geq 0$, as demonstrated by its isoelectronic ion Eu^{3+}).^[35] It is evident that Sm^{2+} can be excited by any wavelength in the 250 – 550 nm UV–visible range. The PL emission spectra (measured under $\lambda_{\text{ex}} = 473\text{ nm}$ radiation) over the 525 – 850 nm spectral and 298 – 648 K temperature ranges are shown in Figure 2b. The chromaticity coordinates of the sample emissions calculated from the PL spectra at two boundary temperatures are given in Figure 2d ((0.568, 0.429) @ 298 K ; (0.561, 0.437) @ 648 K).

In this spectral region, the emission of the coating is featured by one broad-band and a series of sharp-bands. Owing to the low shielding, the outer 5d orbitals strongly interact with ligands, which causes larger crystal field splitting than in the case of the 4f orbitals and broad emission bands in the $4\text{f}5\text{d}$ configuration.^[36] In contrast, the ligands weakly influence the energies of 4f levels that are shielded by the outer-shell 5s and 5p orbitals.^[37] Thus, the broad-band emission emanates from the transitions within the $4\text{f}5\text{d}$ configuration, while the sharp emissions originate from the $4\text{f}4\text{f}$ transitions. In order to clearly present and identify the PL emissions, the spectrum recorded at 498 K was deconvoluted, and its components are shown in Figure 2e. The spectrum consists of the following:

- i) Two broad emissions (red line) that can be modeled by two Gaussians centered around 610 and 645 nm , respectively; these emissions are attributed to $\text{Sm}^{2+} 4\text{f}5\text{d} \rightarrow ^7\text{F}_1$ transitions (abbreviated as $4\text{f}5\text{d} \rightarrow 4\text{f}$ in what follows). The integrated intensity of the broad $4\text{f}5\text{d} \rightarrow 4\text{f}$ emission is weak at room temperature, indicating an efficient cross-relaxation from the $4\text{f}5\text{d}$ configuration to the 4f excited levels of Sm^{2+} .^[38] The emission intensity increases approx. three times with increase in temperature up to 475 K and then decreases at the same rate reaching the starting intensities (Figure 2f).

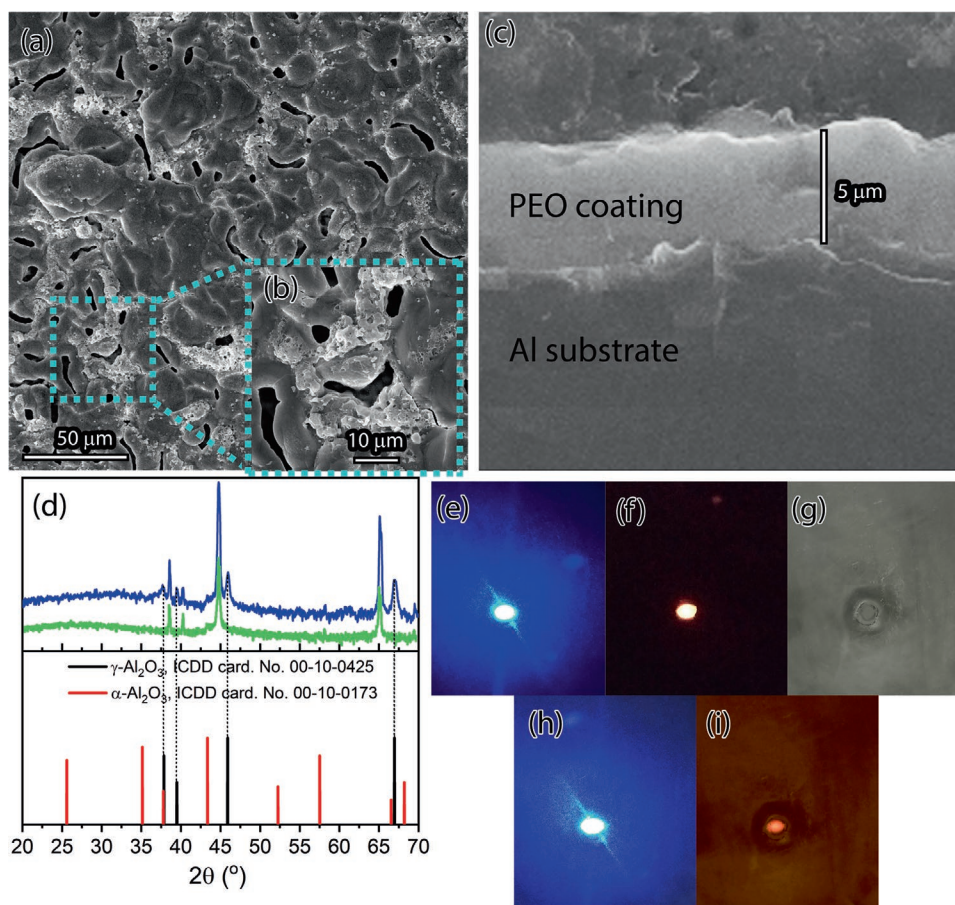


Figure 1. Scanning electron microscopy of the representative aluminum deposited Sm^{2+} -activated $\gamma\text{-Al}_2\text{O}_3$ coating. a) 50 μm bar. b) 10 μm bar top-view. c) Cross-sectional image of the coating (embedded in the commercial epoxy to ease the sample cutting for SEM measurement). d) XRD pattern of the representative Sm^{2+} -doped $\gamma\text{-Al}_2\text{O}_3$ PEO coating deposited on an Al substrate (blue), together with the XRD pattern of the uncoated pure aluminum substrate (green) and corresponding ICDD cards of $\gamma\text{-Al}_2\text{O}_3$ and $\alpha\text{-Al}_2\text{O}_3$. The coating recorded under 473 nm laser irradiation: e, h) in the dark. f, i) With a 550 nm long-pass filter mounted on the camera. g) Under daylight conditions.

- II) Five sharp emission lines (green line) peaking at 688, 702, 729, 768, and 810 nm, which can be attributed to the $\text{Sm}^{2+} {}^5\text{D}_0 \rightarrow {}^7\text{F}_j$ ($j = 0-4$) transitions, respectively (emissions from the ${}^5\text{D}_0 \rightarrow {}^7\text{F}_{5,6}$ transitions were not observed). The phonon side-band to the ${}^5\text{D}_0 \rightarrow {}^7\text{F}_0$ emission, as described in ref. [39], was not observed. The low-intensity emission peak at 678 nm can be attributed to the ${}^5\text{D}_0 \rightarrow {}^7\text{F}_0$ transition at different site symmetries.^[40] This is consistent with the existence of two crystalline sites, which are tetrahedral and octahedral, respectively, that can be occupied by Sm^{2+} in the host $\gamma\text{-Al}_2\text{O}_3$ structure. The intensity of the emission from the ${}^5\text{D}_0$ level uniformly decreased with increasing temperature until almost undetectable (Figure 2f).
- III) Five sharp peaks that overlap with the broad-band emission, peaking at 562, 598, 630, 642, and 665 nm, at higher energies than emissions from the ${}^5\text{D}_0$ level. These peaks have previously been ascribed to the Sm^{3+} impurity in several studies, as they show a resemblance to the Sm^{3+} 4f-4f emissions.^[38,41] However, only the positions of the 562 and 598 nm emissions match the 4f-4f transitions of Sm^{3+} , while the other three emissions have a mismatch of 10 to 15 nm.^[42] In order to clarify the source of the mismatched emissions,

the Sm^{2+} -doped $\gamma\text{-Al}_2\text{O}_3$ coating was illuminated with 450 nm radiation, which is incapable of exciting Sm^{3+} ions. Under 450 nm illumination, the emission peaks at 562 and 598 nm were not observed, while the other three emissions remained, possessing the same relative intensity as other emissions (Figure S1, Supporting Information). Thus, the 562 and 598 nm peaks are ascribed to the ${}^4\text{G}_{5/2} \rightarrow {}^6\text{H}_{5/2}$ and ${}^4\text{G}_{5/2} \rightarrow {}^6\text{H}_{7/2}$ transitions of Sm^{3+} ions (orange line),^[43] respectively, while the remaining three are ascribed to ${}^5\text{D}_1 \rightarrow {}^7\text{F}_j$ ($j = 0, 1, 2$) transitions of Sm^{2+} ions (blue line), as reported in refs. [26,39,44–46]. The intensity of the ${}^5\text{D}_1 \rightarrow {}^7\text{F}_j$ emissions increased $\approx$ three times due to the thermalization from the ${}^5\text{D}_0$ level, reaching a maximum at approximately 400 K and then decreased with increasing temperature until almost undetectable (Figure 2f).

The observed emissions are presented in the energy level diagram in Figure 2g. The emission intensities are proportional to the optical center populations of the emitting levels at a given temperature. The populations depend on the population/depopulation mechanisms depicted by the configuration coordinate diagram in Figure 2h.

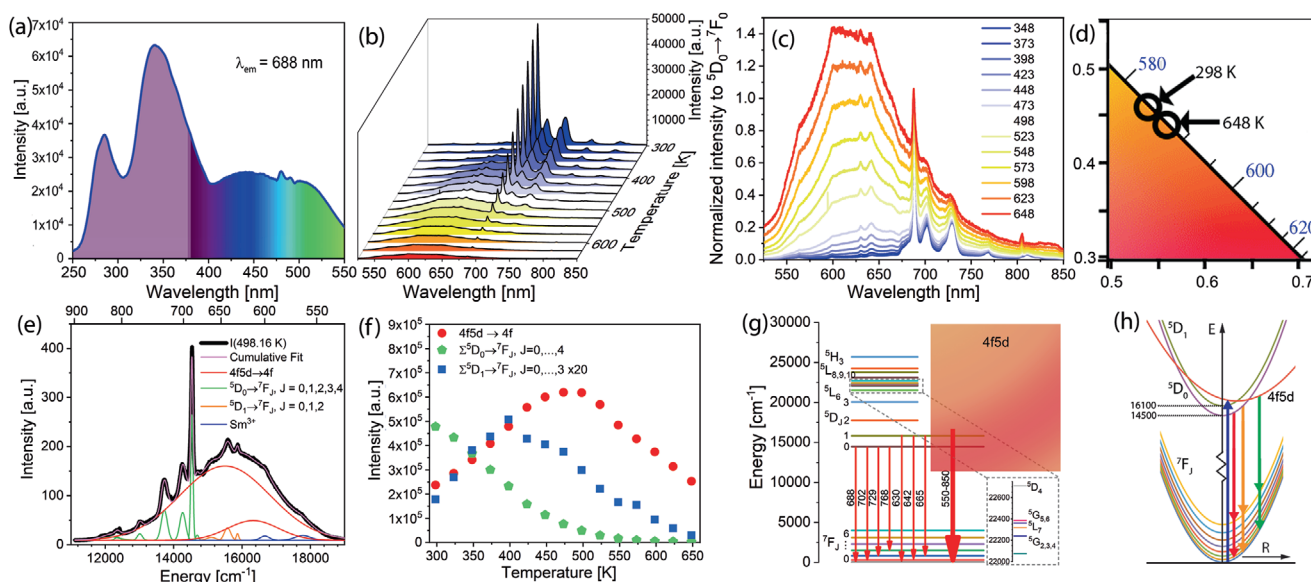


Figure 2. PL of Sm^{2+} -doped $\gamma\text{-Al}_2\text{O}_3$ PEO coating deposited on pure Al substrate. a) PL excitation spectrum (250–550 nm, emission collected at $\lambda_{\text{em}} = 688$ nm). b) Evolution of PL emission spectra (525–850 nm) in the 298–648 K temperature range (excited by 473 nm radiation). c) Normalized PL emission spectra given in Figure 2b. d) Chromaticity coordinates of the sample calculated from PL spectra acquired at two boundary temperatures. e) Deconvolution of the PL emission spectrum measured at 498 K. f) Integrated intensities of $4f5d \rightarrow 4f$, $^5D_0 \rightarrow ^7F_J$, and $^5D_1 \rightarrow ^7F_J$ emissions as a function of temperature. g) Energy level diagram. h) The configuration coordinate diagram.

As already stated, luminescence thermometry is based on temperature-induced changes in the luminescence of a material. The thermometric properties of the PEO coatings were analyzed in detail using three different temperature read-out approaches: i) temperature-induced changes in the emission intensity ratio, ii) the temperature dependence of the luminescence lifetime, and iii) temperature-induced changes in the emission band position (band shift). Figures of merit in the form of parameters that demonstrate the measurement performance, that is, the absolute (S_a) and relative (S_r) sensitivities, and the temperature resolution (ΔT), are given by the following equations:^[13]

$$S_a = \left| \frac{\partial Q}{\partial T} \right| \quad (1)$$

$$S_r = \frac{S_a}{Q} \times 100\% \quad (2)$$

$$\Delta T = \frac{\sigma}{S_a} = \frac{\sigma_r}{S_r} \quad (3)$$

where Q is the luminescence property of interest (the luminescence emission intensity ratio, lifetime, or band-shift), and σ and σ_r are the absolute and relative errors in the determination of Q , respectively. In practice, the resolution depends on the experimental conditions. The absolute sensitivity depends on the measurand, and thus is not suitable for the comparison of the results obtained with different measuring systems. The relative sensitivity resolves this problem as it normalizes absolute sensitivity with respect to Q .^[18]

2.1. Temperature-Induced Changes in the LIR

The LIR temperature read-out is the most popular approach in luminescence thermometry. It is based on the observation of temperature-induced changes in the ratio of two emissions or excitations. It is a simple measurement technique, requires the least complex equipment, and the read-out is not affected by fluctuations in the excitation power. The LIR of two thermally coupled excited levels of the same emission center is given by^[18]

$$\text{LIR} = \frac{I_H}{I_L} = B \exp\left(-\frac{\Delta E}{kT}\right) \quad (4)$$

where I_H is the intensity of the emission from the higher-energy excited state, I_L is the intensity of the emission from the lower-energy excited state, ΔE is the energy difference between the thermalized excited states, k is the Boltzmann constant ($k = 0.695 \text{ cm}^{-1} \text{ K}^{-1}$), $B = \nu_H A_H g_H / \nu_L A_L g_L$, where A is the radiative transition probability, ν is the emission bar-ycenter frequency (or energy), and $g = 2J + 1$ is the level degeneracy. In case that the single photo counting detection is performed, the ν_H / ν_L should be removed for equation for B parameter, as demonstrated by Suta and Meijerink.^[47] This model holds well for the interconfigurational LIR (between 5D_1 and 5D_0 levels) as the multiphonon-deexcitation is the primary depopulation mechanism. However, in the case of LIR between $4f5d$ and 5D_0 , this physical representation requires modification. The intensity of the emission from the 5D_0 level is given by the Mott–Seitz law:

$$I(^5D_0) = \left(1 + C \exp\left(-\frac{E_a}{kT}\right)\right)^{-1} \quad (5)$$

where E_a is an effective activation barrier, that is, the energy difference between the two states of interest, and C is the measure for the strength of the non-radiative thermalization rate normalized to the radiative decay. The non-radiative coupling between 5D_0 and the lowest 4f5d level is proportional to the squared modulus of the respective transition dipole moment. As such a transition is electric dipole allowed it is highly probable, and $C \approx 10^4 \gg 1$. Thus, the Equation (5) approximates to:

$$I(^5D_0) \approx C^{-1} \exp\left(\frac{E_a}{kT}\right) \quad (6)$$

Thus, the LIR between 4f5d and 5D_0 level takes form of the conventional Boltzmann distribution:

$$\text{LIR} = \frac{I(4f5d)}{I(^5D_0)} = C \exp\left(-\frac{E_a}{kT}\right) \quad (7)$$

A thermometric analysis was performed, based on two pairs of emitting levels: i) 4f5d $\rightarrow$ 4f / $^5D_0 \rightarrow ^7F_1$, and ii) $^5D_1 \rightarrow ^7F_1$ / $^5D_0 \rightarrow ^7F_1$. Based on the temperature dependence of the integrated emission intensities presented in Figure 2e, the analysis for pair (i) was performed from 298 to 648 K, and the analysis for pair (ii) was performed from 298 to 548 K. The temperature ranges were chosen according to the visibility of the transitions. For pair (i), the 4f5d $\rightarrow$ 4f transition determined the choice of the lower end of the temperature range, and the $^5D_0 \rightarrow ^7F_1$ transition determined the choice of the upper end. For pair (ii), the choice of both, upper and lower end of the temperature range was determined by the $^5D_1 \rightarrow ^7F_1$ transition. A fit of the experimental data with Equation (4), presented in Figure 3a, showed excellent agreement between experiment and theory (adj. $R^2 = 0.999$ and 0.995).

The absolute and relative sensitivities obtained with the LIR method are presented in Figure 3b and summarized in Table 1. When the 4f5d level is involved in the LIR calculation, the relative sensitivity of the Sm $^{2+}$ -doped $\gamma\text{-Al}_2\text{O}_3$ PEO coating was found to be 4.8% K $^{-1}$ at 298 K. When the 5D_0 and 5D_1 levels of Sm $^{2+}$ ions were involved in the LIR calculation, the relative sensitivity was somewhat lower and found to be 2.6% K $^{-1}$ at 298 K. After 30 consecutive measurements, the obtained temperature resolution was $\Delta T = 0.04$ at 298 K for the 4f5d/ 5D_0 , and $\Delta T = 2$ K at 298 K for the 5D_1 / 5D_0 ratio—a 50 $\times$ better result for the LIR involving 4f5d configuration.

2.2. Temperature-Dependent Luminescence Lifetime

The lifetime of the excited state is the inverse sum of the radiative (k_R) and non-radiative (k_{NR}) transition rates:

$$\tau(T) = \frac{1}{k_R(T) + k_{NR}(T)} = \frac{\tau_R(T)}{1 + \tau_R(T)/\tau_{NR}(T)} \quad (8)$$

where τ_R and τ_{NR} are the radiative and non-radiative lifetimes, respectively.

The lifetime of the 5D_0 level of Sm $^{2+}$ ($\lambda_{ex} = 473$ nm, $\lambda_{em} = 688$ nm) was recorded in the temperature range of 100–620 K (Figure 3c). The temperature dependence of the 5D_0 emission

lifetime is composed of three regions: i) below 300 K, where the lifetime values show a mild decrease with temperature, ii) between 300 and 500 K, where a sharp decline in the lifetime is visible due to the temperature-strengthened non-radiative transitions, and iii) above 500 K, where a mild decrease is visible again.

The temperature dependence of the lifetimes of Lns and TM ions can be described by several models such as the Mott–Seitz model,^[48] the multiphonon model,^[49] the charge-transfer band model,^[50] and the model that accounts for the temperature dependence of the radiative relaxation rate.^[51] These models were tested with experimental data, and the latter provided excellent agreement with the model predictions (Table S1, Supporting Information). This can be justified from a physical perspective also. The $^5D_0 \rightarrow ^7F_0$ transition is usually strongly forbidden and upon vibronic coupling with the energetically close 4f5d1 states, it becomes partially allowed, which makes the decay time decrease at elevated temperatures. At sufficiently high temperatures, crossover occurs, and the decay time fully drops. The temperature dependence of the lifetime when the radiative relaxation rate is considered as temperature-dependent is given by:^[52]

$$\tau(T) = \frac{\tau_0 \cdot \tanh(h\nu/2kT)}{1 + A_{NR} \cdot \tau_0 \cdot \tanh(h\nu/2kT) \cdot \exp(-\Delta E_a/kT)} \quad (9)$$

where τ_0 is the radiative lifetime when the temperature approaches zero, $A_{NR} = 1/\tau_{NR}$ is the non-radiative decay rate, $h\nu$ is the average energy of the phonons involved in the process, and ΔE_a is the activation energy of the process. For the crossover model, the obtained values are $\tau_0 = 1.57$ ms, $A_{NR} = 19\,800$ ms $^{-1}$, $\Delta E_a = 3312$ cm $^{-1}$, and $h\nu = 496$ cm $^{-1}$, with $\chi^2 = 0.027$, indicating a very good fit. The absolute and relative sensitivities obtained for the read-out based on the temperature-dependent luminescence lifetime are presented in Figure 3d. The maximal value of the relative sensitivity is 1.2% K $^{-1}$ @ 540 K, while the estimated temperature resolution for this read-out was found to be $\Delta T = 0.3$ K at 298 K.

2.3. Temperature-Induced Changes in the Emission Band Position

The temperature dependence of the rare-earth ion band-shift is given by the McCumber–Sturge equation:^[53]

$$\Delta\nu = \alpha \left(\frac{T}{T_D}\right)^4 \int_0^{T_D/T} \frac{x^3}{e^x - 1} dx \quad (10)$$

where T_D is the Debye temperature and α is the coupling parameter. However, this equation is only valid for weak electron-phonon coupling, and thus, narrow line emission, that is, it is not suited for describing the broad-band 4f5d-4f emission bandshift. However, for the purpose of calibrating the luminescent thermometer, an approximate equation of an exponential or polynomial form may suffice when the physical interpretation of the previous equation is of no interest.^[54] The choice between those two approximations should be made on the grounds of the simplicity versus fit quality: for practical reasons

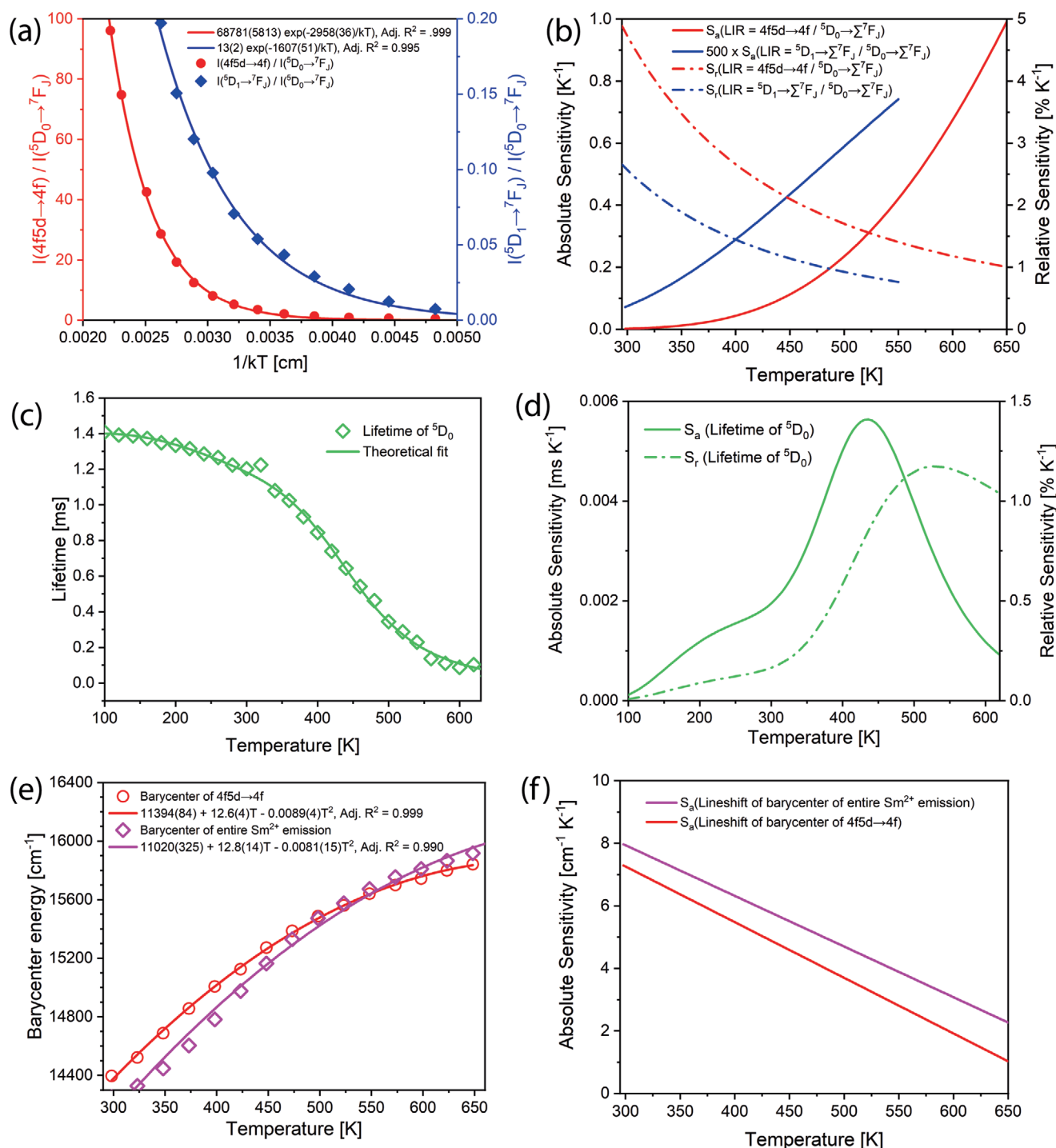


Figure 3. Thermometric properties of the Sm^{2+} -doped $\gamma\text{-Al}_2\text{O}_3$ PEO coating deposited on a pure Al substrate. a) Experimental (symbols) and theoretical (full line) temperature-dependent LIR calculated according to Equation (4) for two pairs of emitting levels: i) $4f5d \rightarrow 4f / {}^5\text{D}_0 \rightarrow {}^7\text{F}_j$ and ii) ${}^5\text{D}_1 \rightarrow {}^7\text{F}_j / {}^5\text{D}_0 \rightarrow {}^7\text{F}_j$. b) Absolute (S_a) and relative (S_r) sensitivities obtained by LIR read-out. c) Experimental (symbols) and theoretical (full line) temperature dependence of the ${}^5\text{D}_0$ level of the Sm^{2+} lifetime values obtained by fitting with the crossover model. d) Absolute (S_a) and relative (S_r) sensitivities obtained by lifetime read-out. e) A fit of the experimental data for $4f5d \rightarrow 4f$ and complete Sm^{2+} emission band barycenter position with a quadratic equation, as an approximation to the McCumber–Sturge equation. f) Absolute (S_a) sensitivity obtained by band-shift read-out.

the simplest approximative equation with an adequate fit quality should be used.

The position of the $4f5d \rightarrow 4f$ transition is estimated from the equation for the barycenter:^[55,56]

Table 1. Comparison of various sensors (most prominent) for luminescence thermometry by LIR, lifetime, and band-shift methods.

Sensor	Form	Method	S_a	S_r [% K ⁻¹]	Temperature range	Ref.
$\gamma\text{-Al}_2\text{O}_3\text{:Sm}^{2+}$	Coating	LIR ^{a)}	1 K ⁻¹ @ 648 K	4.8 @ 298 K	298–648	This work
$\gamma\text{-Al}_2\text{O}_3\text{:Sm}^{2+}$	Coating	LIR ^{b)}	0.0015 K ⁻¹ @ 548 K	2.6 @ 298 K	298–548	This work
$\text{SrB}_4\text{O}_7\text{:Sm}^{2+}$	Powder	LIR (4f5d/ ⁵ D ₀)		2.16 @ 500 K	300–723	[26]
$\text{GdAlO}_3\text{:Eu}^{3+}$	Powder	LIR (⁵ D ₁ / ⁵ D ₀)		2.96 @ 293 K	293–773	[59]
$\gamma\text{-Al}_2\text{O}_3\text{:Sm}^{2+}$	Coating	Lifetime (⁵ D ₀)	0.0056 ms K ⁻¹ @ 436 K	1.2 @ 531 K	100–620	This work
$\text{SrB}_4\text{O}_7\text{:Sm}^{2+}$	Powder	Lifetime		3.36 @ 550 K	300–723	[26]
$\text{GdAlO}_3\text{:Eu}^{3+}$	Powder	Lifetime		2.28 @ 793 K	293–793	[59]
$\text{Li}_{1.8}\text{Na}_{0.2}\text{TiO}_3\text{:Mn}^{4+}$	Powder	Lifetime	0.004 ms K ⁻¹ @ 280 K	2.27 @ 330 K	10–350	[60]
$\gamma\text{-Al}_2\text{O}_3\text{:Sm}^{2+}$	Coating	Band-shift ^{c)}	7.3 cm ⁻¹ K ⁻¹ @ 298 K		298–648	This work
$\gamma\text{-Al}_2\text{O}_3\text{:Sm}^{2+}$	Coating	Band-shift ^{d)}	8 cm ⁻¹ K ⁻¹ @ 298 K		298–648	This work
GaAs	Bulk	Band-shift	3.76 cm ⁻¹ K ⁻¹ @ 300 K		300–600	[61]
$\text{YVO}_4\text{:Dy}^{3+}$	Powder	Band-shift	0.13 cm ⁻¹ K ⁻¹ @ RT		RT–733	[54]
CdSe	Quantum Dot	Band-shift	4.4 cm ⁻¹ K ⁻¹		100–320	[62,63]
CdTe	Quantum Dot	Band-shift	5 cm ⁻¹ K ⁻¹		300–353	[64]
Ag ₂ S	Quantum Dot	Band-shift	11 cm ⁻¹ K ⁻¹		295–317	[65]
$\text{YVO}_4\text{:Eu}^{3+}$	Bulk	Band-shift	0.20 cm ⁻¹ K ⁻¹ @ RT		293–733	[54]
$\text{Y}_2\text{O}_3\text{:Eu}^{3+}$	Powder	Band-shift	0.04 cm ⁻¹ K ⁻¹		257–345	[66]
$\text{SrTiO}_3\text{:Eu}^{3+}$	Powder	Band-shift	0.03 cm ⁻¹ K ⁻¹		4–420	[67]

^{a)}4f5d/⁵D₀; ^{b)}⁵D₁/⁵D₀; ^{c)}Barycenter of the 4f5d emission; ^{d)}Barycenter of the complete Sm²⁺ emission band.

$$\tilde{\nu}(\text{cm}^{-1}) = \int \tilde{\nu} i(\text{cm}^{-1}) d\tilde{\nu} / \int i(\text{cm}^{-1}) d\tilde{\nu} \quad (11)$$

where i (cm⁻¹) is the intensity obtained by the Jacobean transformation of the spectra recorded in nanometers by:^[57]

$$i(\text{cm}^{-1}) = \lambda^2 i(\text{nm}) / 10^7 \quad (12)$$

The fitting of the experimental data for the 4f5d→4f transition and the barycenter position of the entire Sm²⁺ emission as obtained with Equation (7) are presented in Figure 3e, showing excellent agreement between experiment and theory (adj. R^2 = 0.999 and 0.990). The emission from the 4f5d configuration is significantly blue-shifted in energy by ≈1600 cm⁻¹ over the temperature range 298–648 K. At higher temperature the lattice expands, that is, the Sm–O distances increase on average, thus decreasing the crystal field splitting. Consequently, the emission energy has to increase since the lowest 4f5d excited state lies at higher energies. Conversely, as the 4f orbitals are shielded by the outer electrons their inter-configurational transitions are weakly affected by the crystal field making their band shift almost negligible in comparison.^[58]

The absolute sensitivities were found to be 7.3 cm⁻¹ K⁻¹ at 298 K when only the 4f5d emission barycenters were analyzed, and 8 cm⁻¹ K⁻¹ at 298 K for the complete Sm²⁺ emission (Figure 3f). The temperature resolution of this read-out is $\Delta T = 0.06$ K @ 298 K.

2.4. Performance Comparison

The absolute and relative sensitivities obtained with different read-out methods are summarized in Table 1 and compared

with other published data. All the approaches resulted in highly sensitive coatings that could be adopted for versatile temperature read-out setups. The relative sensitivity of 4.8% K⁻¹ at 298 K for the 4f5d LIR read-out was higher than that of the most sensitive Ln, its isoelectronic Eu³⁺ ion.^[59] This was to be expected because the relative sensitivity is directly proportional to the energy gap, which is larger between the 4f5d and ⁵D₀ levels in Sm²⁺ ions than between the ⁵D₁ and ⁵D₀ levels in Eu³⁺ ions (≈1750 cm⁻¹). The relative sensitivity of 2.6% K⁻¹ at 298 K for the ⁵D₀ / ⁵D₁ LIR read-out was lower than the LIR of the same levels of Eu³⁺ ions. However, the ⁵D₁ emission intensity in Eu³⁺ ions is usually very weak, so the LIR method involving the ⁵D_{1,0} levels of Sm²⁺ provided a better temperature resolution because of the more intense ⁵D₁ emission.

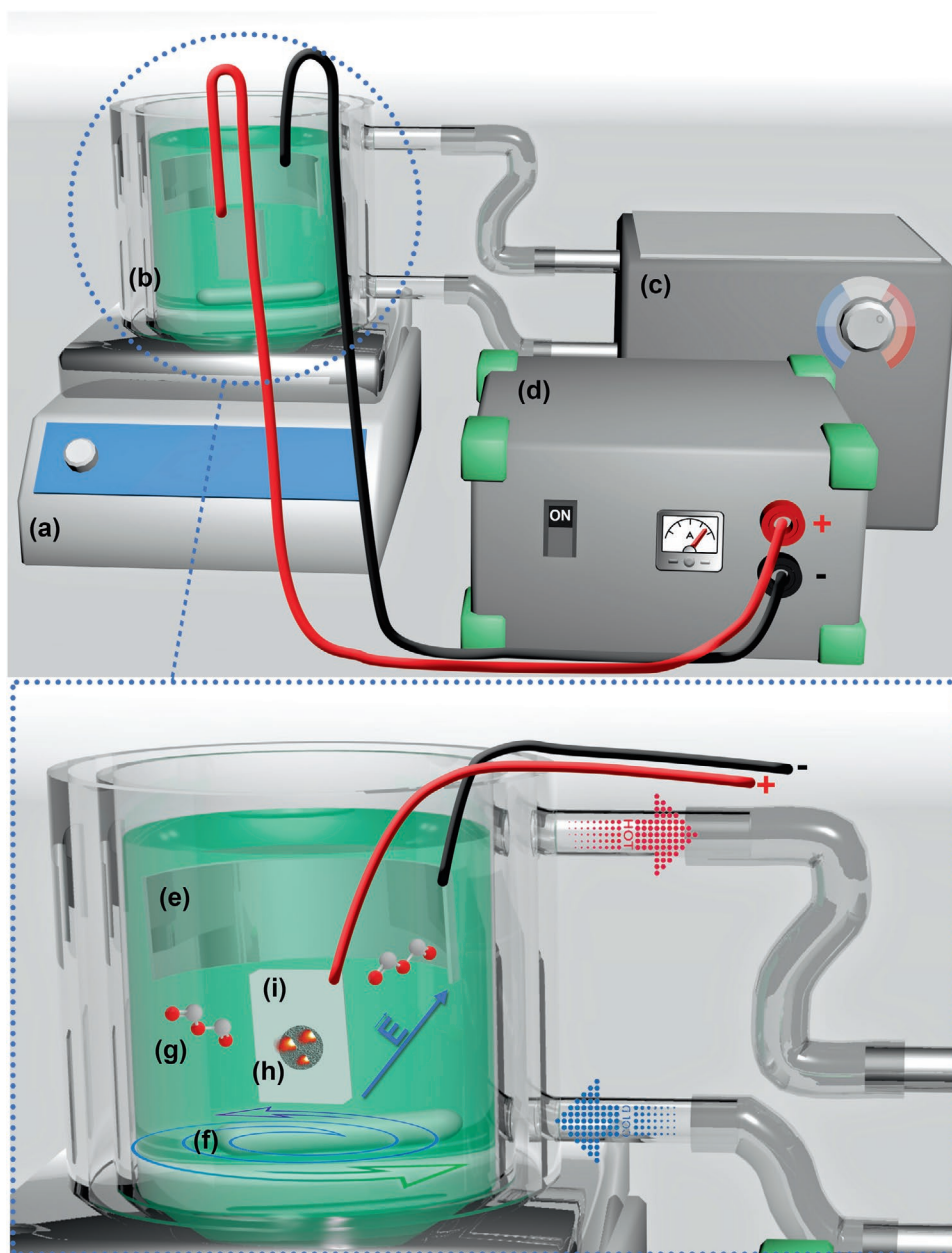
For the temperature-dependent luminescence lifetime read-out, Sm²⁺-doped $\gamma\text{-Al}_2\text{O}_3$ coatings showed a relative sensitivity of 1.2% K⁻¹ at 540 K. Although this value was still sufficiently high for precise optical temperature measurements, it was lower than that of Sm²⁺-doped SrB₄O₇ or Mn⁴⁺ and Eu³⁺ ions in different hosts (see Table 1).

The sensitivity of the temperature read-out from the emission band position was significantly better than those of all Ln-based luminescence thermometers and the commercial semiconductor GaAs probe, while it was comparable to those of the most perspective quantum dot probes, as summarized in Table 1.

Considering that this thermometric coating enables multiparametric thermal sensing, it is possible to converge the different thermal read-outs to eliminate potential systematic errors in measurements and to achieve high measurement reliability, as recently demonstrated by Shen et al.^[65]

One of the potential limitations of the proposed study is the fact that Ln^{2+} may oxidize to Ln^{3+} . However, this occurs at temperatures higher than 500 °C (≈ 773 K),^[68,69] while the temperature range of the proposed luminescence thermometer is up to 375 °C (≈ 648 K). Also, during thermometry experiments samples were heated and cooled numerous times and no change in the emission properties is observed. So far, control of the concentration of incorporated Sm^{3+} was unfortunately not possible, although it is demonstrated that Sm^{3+} emission is very low and can be easily eliminated by the appropriate excitation, for example, 450 nm. Improvement of the coating preparation technique to eliminate Sm^{3+} is underway in our lab. On the

other hand, considerable advantage of the proposed deposition method is the fact that the application of PEO is limited only by the size of the electrolytic cell so, if the substrate fits within the bath, the coatings will be uniformly grown on it, regardless its shape and size. Therefore, it is possible to extend the film deposition on large-scale metal substrates of arbitrary shape. Also, as a good alternative to ruby, narrow 0-0 emission line of Sm^{2+} is often used for high-pressure sensing.^[70–72] Therefore, interesting application that can be exploited in the future is pressure sensing ability of the coatings. In addition, as demonstrated recently, such luminescence-based thermometers can be used as optical vacuum sensors also.^[73]



Scheme 1. The experimental setup for the PEO process. a) Magnetic stirrer. b) Double-wall electrolytic cell. c) Temperature-controlled water pump/bath. d) Constant-current power supply. e) Stainless steel electrode. f) Spinning bar. g) Sm_2O_3 particles. h) Anode—aluminum surface exposed to the electrolyte, with micro discharges appearing during the process. i) Insulation resin.

3. Conclusion

Sm²⁺-doped γ -Al₂O₃ coatings were successfully synthesized by the PEO method on pure aluminum and on 6061 and 7075 alloy substrates, adding new functionality to the diversity of aluminum features which have been exploited previously. The coatings can be optically excited by any source in the UV-visible range (250–550 nm) to produce strong emission. The emission spectra ($\lambda_{\text{ex}} = 473$ nm, $T = 298$ –648 K) are characterized by Sm²⁺ emissions from electronic transitions of both the 4f5d and the 4f levels to the ⁷F_J ground multiplet, showing different temperature dependences. The thermometric properties of the coatings were analyzed in detail using three different read-out approaches: i) temperature-induced changes in emission intensities, ii) temperature-dependent luminescence lifetimes, and iii) temperature-induced changes in the emission band position. The LIR method involving the 4f5d-4f and ⁵D₀→⁷F_J transitions of a single Sm²⁺ emission center led to a very high relative sensitivity of $S_r \approx 4.8\%$ K⁻¹ at 298 K and an estimated temperature resolution of 0.04 K. The lifetimes of the Sm²⁺ ions ⁵D₀ level were acquired in the temperature range of 100–620 K and fitted to the model accounting temperature dependence of the radiative lifetime. The fit provided a value for the maximal relative sensitivity of $S_r \approx 1.2\%$ K⁻¹ at 540 K, with a temperature resolution of 0.3 K at 298 K. The barycenter of the Sm²⁺ ions for the 4f5d-4f emission was significantly blue-shifted to higher energies in the measured range of 298–648 K, leading to the highest sensitivity observed so far with Ln-activated probes, that is, 8 cm⁻¹ K⁻¹ @ 298 K, with an estimated temperature resolution of 0.06 K. For the commercially available GaAs probe, the manufacturer claims twice the sensitivity of 3.76 cm⁻¹ K⁻¹ and a temperature resolution of 0.1 K at 298 K.

With respect to the versatility of possible temperature read-outs and the simplicity of coating deposition, future work should proceed in two directions: i) the testing of thermometric coatings for temperature measurements on nuclear waste containers and as temperature sensors for the aviation and automotive industries, and ii) the construction of a portable coating device capable of mobile and off-site sensor deposition.

4. Experimental Section

Three substrates: 99.9% pure aluminum foil and the 6061 and 7075 commercial aluminum alloys, respectively, were used as anodes during the PEO process. The experimental setup for the PEO process is shown in **Scheme 1** and described in detail in ref.^[74] The pre-processing included degreasing in acetone, ultrasonic cleansing, and subsequent drying in the warm airstream. The relevant part of the foil's surface was masked with insulation resin to create a circular active surface for oxidation (see Scheme 1). The samples were immersed in a water solution of 0.1 M H₃BO₃ (Sigma Aldrich) and 0.05 M Na₂B₄O₇·10H₂O (Sigma Aldrich), with the addition of 4 g L⁻¹ Sm₂O₃ (99.9%, Pangea International). The amount of the samarium precursor was already optimized in the previous work.^[21] The process was run for 10 min under a constant current density of 150 mA cm⁻². The temperature of the electrolyte was kept constant at 18 °C. The postprocessing included rinsing in distilled water to prevent the additional deposition of electrolyte constituents.

Scanning electron microscopy (SEM, MIRA-3 TESCAN) was used for the morphological characterization of the created coatings. XRD was used to investigate the coating crystallinity (Rigaku Ultima IV diffractometer with Ni-filtered CuK α radiation ($\lambda = 1.54178$ Å)). The

samples were scanned in the Bragg-Brentano geometry, with a 2° min⁻¹ acquisition rate and a step size of 0.02°. PL excitation spectra were obtained using a Horiba Jobin Yvon Fluorolog FL3-22 spectrofluorometer equipped with a 450 W Xenon lamp as an excitation source. For acquisition of the emission spectra, a high-stability 473 nm laser with an optical power of 150 mW was used. The excitation and emission beams were transmitted via a fiber-optic bundle. The emission spectra were recorded using a high-resolution spectrograph (FHR 1000) with an ICCD detector (Horiba Jobin-Yvon). The samples were placed on the liquid nitrogen-cooled hot/cold stage with a controllable temperature range of 77–750 K and a temperature resolution of 0.05 K. Additionally, a 450 nm laser was used for the separation of Sm³⁺ from the Sm²⁺ 4f-4f peaks. Luminescence spectra were not corrected for the apparatus response. The temperature resolutions were calculated from the 30 spectra recorded under the same conditions.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

coatings, luminescence, Sm²⁺, thermometry, γ -Al₂O₃

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